

RedBook AI Assistant

Domain-Specific RAG System for UK Traffic Management (NRSWA 1991)

Developer Portfolio & Technical Architecture Overview | Augustine Kodom

1. Executive Summary & Project Purpose

RedBook AI is a production-grade, domain-specific AI assistant and decision-support web application engineered specifically for UK Traffic Management (TM) Operatives, Site Supervisors, and NRSWA Safety Inspectors.

The system is strictly grounded in the Department for Transport (DfT) statutory Code of Practice: "Safety at Street Works and Road Works", backed by Section 65 of the New Roads and Street Works Act 1991 (NRSWA). It solves a critical industry challenge: delivering instant, 100% accurate, zero-hallucination safety guidance to operatives working in high-risk outdoor roadwork environments.

2. Key Problem & Engineering Solution

The Industry Problem

- Roadwork operatives must comply with strict legal safety zone clearances on site.
- Consulting physical 100+ page manuals in harsh outdoor weather is slow and error-prone.
- Incorrect safety zone setups result in statutory fines, legal liability, and physical danger.

The Engineering Solution

- Grounded RAG AI Engine linking questions directly to exact Red Book Page & Table citations.
- Mobile-first UI with pitch-black theme & fixed bottom navigation dock for thumb-reachability.
- Interactive Calculators for instant speed limit -> safety zone & taper calculations.
- 100% Zero-Cost Architecture (\$0/month operating cost).

3. Core Application Features & UX Innovations

• Bold Definitive Top Answer Summary:

To prevent operatives from scanning long paragraphs on site, answers begin with a bold, direct 1-sentence summary callout box before detailing statutory code breakdowns.

• Progressive Typewriter Streaming Reveal:

Implemented custom character-by-character streaming animation with a glowing cursor, delivering a conversational AI experience without layout dumping.

• Mobile View Optimization & Bottom Navigation Dock:

Designed an uncluttered top header paired with a glassmorphic bottom navigation tab bar (AI Chat, Calculators, Code Browser) for seamless mobile usage.

• Touch-Optimized Interactive Safety Calculators:

Includes Table 1 Safety Zone & Lead-in Taper Calculator, Traffic Control Selector (Give & Take / Priority / Portable Signals), and Pedestrian Footway Clearance Evaluator.

• Hands-Free Web Speech API Voice Input:

Integrated browser-native Speech-to-Text allowing operatives holding equipment or wearing work gloves to ask questions

hands-free.

4. Technical Architecture & Tech Stack

Layer	Technology / Library	Engineering Rationale
Frontend Framework	React 18 + Vite 4	Fast single-page app architecture with 10s build times.
UI & Styling	Pitch Black Theme + Tailwind CSS	High-contrast outdoor obsidian theme with touch targets >= 44px.
Icons & Design	Lucide React	Consistent, modern vector icons for domain controls.
AI & RAG Pipeline	@google/generative-ai + BM25	Gemini 1.5 Flash integration + standalone local vector engine.
Database / Storage	PostgreSQL + pgvector / Local KB	Grounded text chunks with HNSW vector similarity search.
Speech Recognition	Web Speech API (Browser)	Zero-dependency native speech-to-text for hands-free voice input.
Deployment & CI/CD	Vercel Serverless + GitHub	Automatic production deployments with \$0 operating cost.

5. Key Software Engineering Competencies Demonstrated

1. Full-Stack Web Development: Modular React architecture, pitch-black responsive design, custom hooks, and state management.
2. AI / RAG Engineering: Structuring un-structured PDF manuals into vectorized knowledge bases, prompt guardrails, and citation tracking.
3. Product & UX Design: User-centered design for field operatives (hands-free mic, mobile bottom dock, typewriter reveal).
4. Cost Optimization: Engineered a \$0/month architecture leveraging serverless Vercel hosting and free-tier APIs.
5. Production Readiness: Clean CI/CD pipeline, Git version control, Vercel SPA route rewrites, and environment variable configuration.

6. Repository & Live Links

- **GitHub Repository:** <https://github.com/Augustinekodom/redbook-tm-assistant>
- **Live Demo URL:** <https://redbook-tm-assistant.vercel.app>

